

Module 2 Lesson 1

Arithmetic and Geometric Sequences (5.7)

Objectives

In this lesson, we will learn how to

- write terms of an arithmetic sequence;
- use the formula for the general term of an arithmetic sequence;
- write terms of a geometric sequence;
- use the formula for the general term of a geometric sequence.

1. Sequences

A **sequence** is a list of numbers that are related to each other by a rule.

The numbers in a sequence are called its **terms**.

For example,

$$2, 4, 6, 8, 10, \dots$$

is a sequence because each term is obtained by adding 2 to the previous term.

Terms of a Sequence

The first term is usually written as

$$a_1.$$

The second term is written as

$$a_2.$$

The n th term is written as

$$a_n.$$

A famous sequence is the Fibonacci sequence:

$$1, 1, 2, 3, 5, 8, 13, \dots$$

Each term after the first two is found by adding the two previous terms:

$$1 + 1 = 2,$$

$$1 + 2 = 3,$$

$$2 + 3 = 5,$$

$$3 + 5 = 8.$$

2. Arithmetic Sequences

An arithmetic sequence is a sequence in which each term after the first differs from the preceding term by a constant amount.

This constant amount is called the common difference.

The common difference is usually represented by d .

Arithmetic Sequence

A sequence is arithmetic if the difference between consecutive terms is constant.

$$d = a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = \cdots$$

Examples:

142, 146, 150, 154, 158, ...

Here,

$$d = 146 - 142 = 4.$$

-5, -2, 1, 4, 7, ...

Here,

$$d = -2 - (-5) = 3.$$

8, 3, -2, -7, -12, ...

Here,

$$d = 3 - 8 = -5.$$

3. Writing Terms of an Arithmetic Sequence

Example 1: Write the first six terms of the arithmetic sequence with first term 6 and common difference 4.

Solution

The first term is

$$6.$$

The second term is

$$6 + 4 = 10.$$

The third term is

$$10 + 4 = 14.$$

Continuing this pattern, the first six terms are

$$6, 10, 14, 18, 22, 26.$$

4. General Term of an Arithmetic Sequence

Consider an arithmetic sequence with first term a_1 and common difference d .
The terms follow this pattern:

$$a_1,$$

$$a_1 + d,$$

$$a_1 + 2d,$$

$$a_1 + 3d,$$

and so on.

General Term of an Arithmetic Sequence

The n th term of an arithmetic sequence with first term a_1 and common difference d is

$$a_n = a_1 + (n - 1)d.$$

Example 2: Find the eighth term of the arithmetic sequence whose first term is 4 and whose common difference is -7 .

Solution

Use the formula

$$a_n = a_1 + (n - 1)d.$$

Here,

$$a_1 = 4, \quad d = -7, \quad n = 8.$$

$$a_8 = 4 + (8 - 1)(-7)$$

$$= 4 + 7(-7)$$

$$= 4 - 49$$

$$= -45.$$

The eighth term is

$$-45.$$

5. Geometric Sequences

A **geometric sequence** is a sequence in which each term after the first is obtained by multiplying the preceding term by a fixed nonzero constant.

This fixed constant is called the **common ratio**.

The common ratio is usually represented by r .

Geometric Sequence

A sequence is geometric if the ratio of consecutive terms is constant.

$$r = \frac{a_2}{a_1} = \frac{a_3}{a_2} = \frac{a_4}{a_3} = \dots$$

Examples:

$$1, 5, 25, 125, 625, \dots$$

Here,

$$r = 5.$$

$$4, 8, 16, 32, 64, \dots$$

Here,

$$r = 2.$$

$$6, -12, 24, -48, 96, \dots$$

Here,

$$r = -2.$$

6. Writing Terms of a Geometric Sequence

Example 3: Write the first six terms of the geometric sequence with first term 6 and common ratio

$$\frac{1}{3}.$$

Solution

The first term is

$$6.$$

The second term is

$$6 \cdot \frac{1}{3} = 2.$$

The third term is

$$2 \cdot \frac{1}{3} = \frac{2}{3}.$$

The fourth term is

$$\frac{2}{3} \cdot \frac{1}{3} = \frac{2}{9}.$$

The fifth term is

$$\frac{2}{9} \cdot \frac{1}{3} = \frac{2}{27}.$$

The sixth term is

$$\frac{2}{27} \cdot \frac{1}{3} = \frac{2}{81}.$$

The first six terms are

$$6, 2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \frac{2}{81}.$$

7. General Term of a Geometric Sequence

Consider a geometric sequence with first term a_1 and common ratio r .
The terms follow this pattern:

$$a_1,$$

$$a_1r,$$

$$a_1r^2,$$

$$a_1r^3,$$

and so on.

General Term of a Geometric Sequence

The n th term of a geometric sequence with first term a_1 and common ratio r is

$$a_n = a_1r^{n-1}.$$

8. Using the General Term Formula

Example 4: Find the eighth term in the geometric sequence whose first term is -4 and whose common ratio is -2 .

Solution

Use the formula

$$a_n = a_1 r^{n-1}.$$

Here,

$$a_1 = -4, \quad r = -2, \quad n = 8.$$

$$a_8 = -4(-2)^{8-1}$$

$$= -4(-2)^7$$

$$= -4(-128)$$

$$= 512.$$

The eighth term is

$$512.$$

9. Arithmetic vs. Geometric Sequences

How to Tell the Difference

- In an arithmetic sequence, we add the same number each time.
- In a geometric sequence, we multiply by the same nonzero number each time.

Type of Sequence	What Stays Constant?	Formula
Arithmetic	Common difference d	$a_n = a_1 + (n - 1)d$
Geometric	Common ratio r	$a_n = a_1 r^{n-1}$

Summary

Key Ideas

- A sequence is a list of numbers related by a rule.
- The numbers in a sequence are called terms.
- An arithmetic sequence has a constant common difference.
- The general term of an arithmetic sequence is

$$a_n = a_1 + (n - 1)d.$$

- A geometric sequence has a constant common ratio.
- The general term of a geometric sequence is

$$a_n = a_1 r^{n-1}.$$

- Arithmetic sequences use repeated addition.
- Geometric sequences use repeated multiplication.